

A 0D pulsatile model of aortic-arch banding

Reproducing carotid pulsatility redistribution (Eberth et al., 2009)

svZeroDSolver

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The problem

Eberth et al. (2009): a stiff band on the mouse aortic arch, placed *between* the two carotid takeoffs, creates paired carotids with **similar mean pressure & flow but very different pulsatility**.

- **RCCA-B** (right) — *upstream* of the band: sees the full pulse + reflections $\Rightarrow$ pulsatility index rises ($PI \approx 3.11$).
- **LCCA-B** (left) — *downstream*: the resistive–inertial band damps the pulse ($PI \approx 1.65$); baseline CCA ≈ 1.16 .

Goal: a reduced-order (0D) lumped model that reproduces this redistribution with the band as the only structural change — and that maps to the wall growth & remodeling Eberth measured.

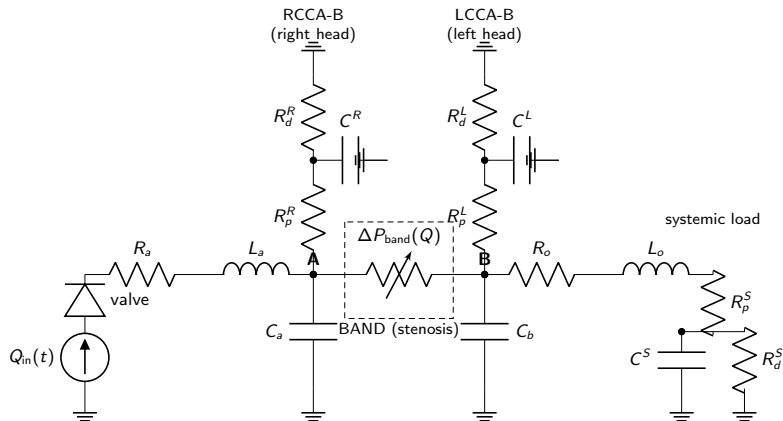
The 0D model

Heart $\rightarrow$ aortic valve $\rightarrow$ ascending aorta (R_a, L_a, C_a) $\rightarrow$ **node A** $\rightarrow$ band $\Delta P_{\text{band}}(Q) \rightarrow$ **node B** $\rightarrow$ descending aorta (R_o, L_o, C_b); a carotid RCR bed (R_p, C, R_d) hangs off each node.

- Vessels: RCL BloodVessel blocks
- Band: nonlinear *Young-Tsai* stenosis
$$\Delta P_{\text{band}} = (R_{\text{band}} + S|Q|)Q + L_{\text{band}} \dot{Q}$$
- Terminal beds: 3-element RCR windkessels
- Valve: ValveTanh diode
- Parameters from mouse literature (geometry, blood, wall stiffness E)
- Every parameter tagged
FIXED / DERIVED / CALIBRATE
- Only the band `stenosis_coefficient` is calibrated (to the Table-1 PI ratio)

Three states differ *only* in carotid geometry: pre-banding (baseline), acute (baseline walls), chronic (Table-1 remodeled walls).

0D circuit (electric analogy)

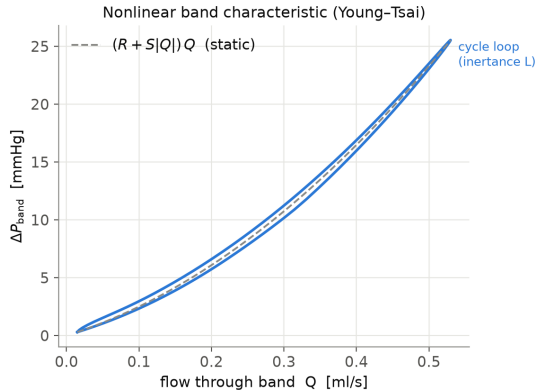


symbol	component (this repo)
$Q_{in}(t)$	INFLOW
valve	aortic_valve
R_a, L_a, C_a	ascending_aorta
$\Delta P_{band}(Q)$	aortic_band
R_o, L_o, C_b	descending_aorta
node A / B	node_A / node_B
R_p^R, C^R, R_d^R	RCR_RIGHT
R_p^L, C^L, R_d^L	RCR_LEFT
R_p^S, C^S, R_d^S	RCR_SYS

Not drawn (negligible): aortic_root (R, L before the valve) and the carotid conduits `rcca.b/lcca.b` (R, L, C between node A/B and `RCR_RIGHT/RCR_LEFT`) — their own wall compliance is tiny next to the bed/aortic C .

Pressure $\leftrightarrow$ voltage, flow $\leftrightarrow$ current. Node A (RCCA-B) is upstream of the band and stays pulsatile; node B (LCCA-B) is downstream and damped. Each terminal bed is an R_p - C - R_d Windkessel.

The band is a nonlinear (Young–Tsai) element



ΔP – Q across the band: the quadratic Bernoulli/turbulent term ($S|Q|Q$) plus an inertance loop ($L \dot{Q}$). This nonlinearity + inertia is what damps the downstream pulse while leaving the upstream node pulsatile.

Parameter roles: fixed, assumed, and the knob we tune

Fixed (measured / literature)

- blood $\rho = 1.06$, $\mu = 3.5$ cP
- wall E : aorta 1.0, carotid 1.5 MPa
- HR 6.09 Hz, CO 0.20 ml/s, MAP 90
- carotid ID/wall (Table 1), aorta dia.
- measured central aortic compliance

⇒ all vessel R , L , C are *derived* from these via Eq. (1)–(3).

Assumed (plausible, not fitted)

- segment lengths (carotid, band) — estimates
- ejection waveform shape (systolic fraction)
- RCR split $R_p:R_d = 10:90$; time const. τ
- distal pressure $P_d = 0$
- valve R_{max}/R_{min} /steepness

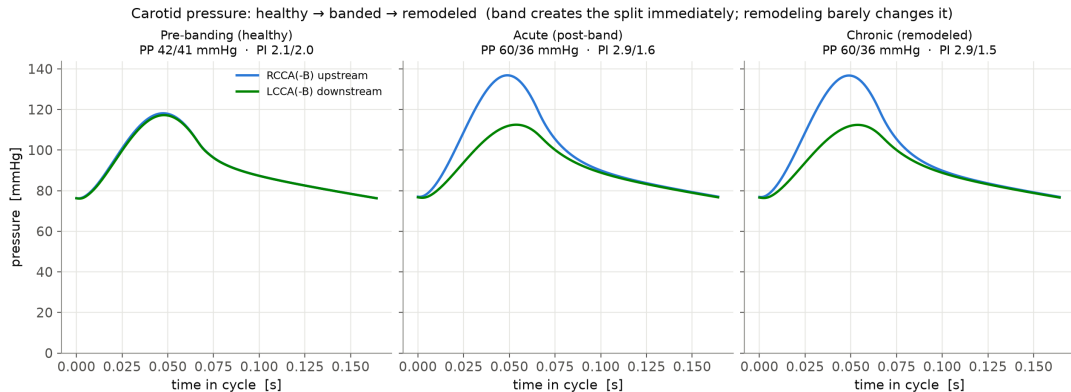
chosen by convention/estimate — could be refined, but not tuned to the data.

Knob (calibrated to data)

- band stenosis_coefficient S
- the **one** parameter we turn
- geometry gives a prior; tuned to the Table-1 PI ratio (3.11/1.65)

Everything else is fixed or derived — this is the only free knob fitted to the data.

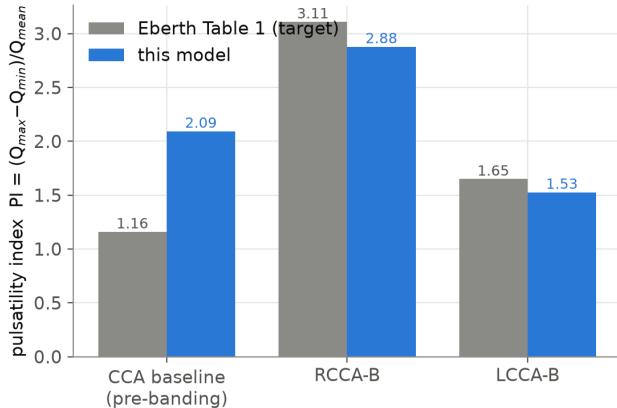
Result: banding creates the split immediately



Carotid pressure is **symmetric before banding**, splits **immediately** on banding (upstream $\uparrow$, downstream damped), and **barely changes** with wall remodeling $\Rightarrow$ the split is set by band *position*; remodeling is the response, not the cause.

Result: pulsatility vs. the paper

Pulsatility by location — calibrated model vs paper Table 1
(stenosis calibrated to the PI ratio; both PIs within ~7%)



	Paper	Model
RCCA-B PI	3.11	≈ 2.9
LCCA-B PI	1.65	≈ 1.5
PI ratio	1.89	1.89
PP RCCA-B	56	60
MAP (R/L)	similar	98/91

The **banded** carotids, the PI ratio, MAP similarity, and pulse pressure are reproduced; mean carotid flows land in the measured range.

Limitation: baseline vs. banded cannot both be matched in 0D

- The model **over-predicts the baseline (pre-banding) PI** (≈ 2.1 vs. the paper's 1.16), though baseline *pulse pressure* matches (≈ 41 vs. 42).
- A parameter sweep shows every knob that lowers the baseline PI to 1.16 **also collapses the banded PI** — the passive bed is a linear pulsatility filter.
- The paper's baseline→banded amplification is $\sim 2.7\times$ ($1.16\rightarrow 3.11$); a lumped 0D band delivers only $\sim 1.5\times$.
- The missing $\sim 2\times$ is **wave reflection** off the band — a spatial/timing effect a 0D element cannot represent.

⇒ Reproducing the *full* amplitude needs a **1D** (wave-propagation) model — exactly the 0D→1D boundary.

Targets and knobs, by group

Control group (CCA) — 4 targets

target (measured)	knob that sets it
$\bar{Q} = 0.016$ ml/s	$R_p^{R,L} + R_d^{R,L}$ (RCR_RIGHT/LEFT)
MAP = 92 mmHg	$R_p^S + R_d^S$ (RCR_SYS)
PP = 42 mmHg	C_a, C_b (aortic)
PI = 1.16	$C^{R,L}$ (RCR_RIGHT/LEFT)

Banded group (RCCA-B / LCCA-B) — 8 targets

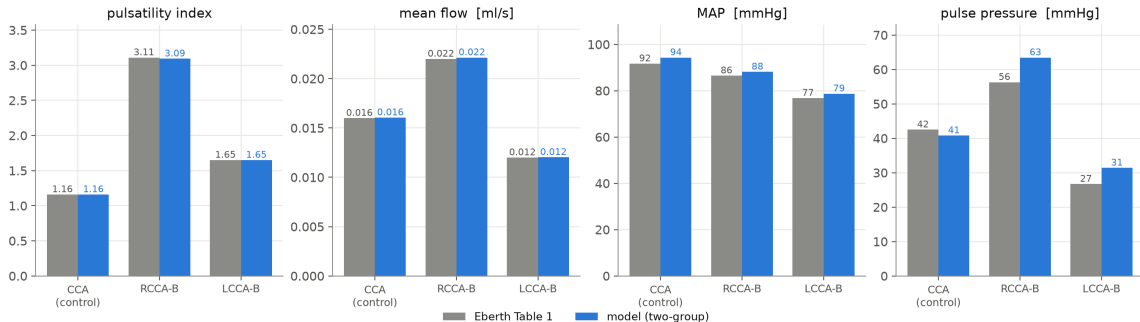
target (measured)	knob that sets it
$\bar{Q} = 0.022 / 0.012$	$R_p^R + R_d^R / R_p^L + R_d^L$
MAP = 86 / 77	S (A→B drop) + $R_p^S + R_d^S$
PP = 56 / 27	C_a, C_b (aortic)
PI = 3.11 / 1.65	C^R / C^L

Bed resistances ($R_p + R_d$) are *pinned* by $(\bar{Q}, \text{MAP})$; the actively-tuned knobs are the bed compliances C^R, C^L ($\rightarrow$ PI) and, for banded, the band S ($\rightarrow$ the A→B pressure drop). Fixed for both groups: HR (7.17 / 6.09), cardiac output CO, carotid geometry, blood μ, ρ , wall moduli E , ejection shape.

Revised: two-group calibration (control vs. banded)

Control (CCA) and banded (RCCA-B/LCCA-B) are **two separate animal groups** — they need not share $R/C/L$. Calibrating each on its own terms — HR and measured flows/pressures pin the beds, the band anchored to the measured $A \rightarrow B$ pressure drop, per-carotid bed compliance sets the PI — matches *all three* carotid states with a stock RCR:

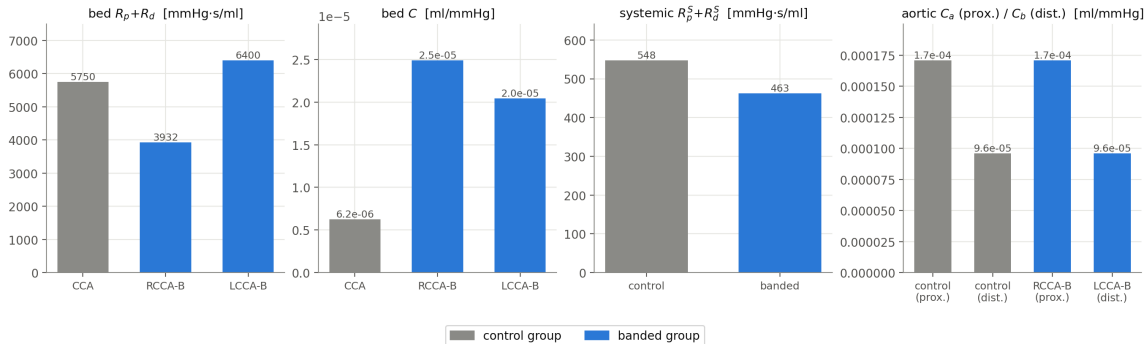
Two-group calibration vs Eberth Table 1 (control CCA + banded RCCA-B/LCCA-B)



DOF-balanced (12 targets / 12 knobs). Cost: bed compliance C^R, C^L differs between groups (fitted, not measured) — descriptive, not a single predictive model.

Settled calibration knobs: control vs. banded

Settled calibration knobs (R_p+R_d and C ; band S excluded): control vs banded



R_p+R_d (bed R) sets the flow split (RCCA-B low $\rightarrow$ more flow); the banded bed *compliance* C^R, C^L is $\sim 3-4\times$ the control's $C^{R,L}$ — a **fitted** difference, not measured (opposite to what autoregulation would do). Systemic $R_p^S+R_d^S$ is essentially unchanged; aortic C_a, C_b (proximal/distal, either side of the band) is **not tuned in either group** — same 0.64/0.36 split of the measured central aortic compliance for control and banded alike.

Summary

- A literature-parameterized 0D model reproduces the Eberth **pulsatility redistribution**: upstream carotid pulsatile, downstream damped, at matched mean pressure/flow.
- Banding creates the split **immediately**; wall remodeling barely changes it — consistent with pulsatility being the *stimulus* for remodeling.
- With a *single shared* parameter set, 0D matches the redistribution and pressures but not both the baseline and banded PIs (a bounded amplification limit; a 1D wave-reflection effect is small for the mouse).
- Treating control and banded as *separate groups* — their own beds, band anchored to the measured pressure drop — matches **all three carotid states**, at the cost of group-specific (fitted) bed compliance.